

FFAR Final Progress Report

As part of closing out a FFAR grant, grantees must complete a Final Progress Report. Please use the template below to complete the programmatic report. The report must be submitted to FFAR within 90 days after the expiration or termination of a FFAR funded research grant. All questions about this form should be directed to grants@foundationfar.org.

The Final Progress Report communicates the cumulative results and major accomplishments of the funded grant research, including accomplishment for each aim and whether all research aims were fully completed. FFAR uses the content of the Final Progress Report to inform Congress, the Advisory Board, and other stakeholders on the successes, impacts and value of funding unique partnerships to support innovative science addressing today's food and agriculture challenges. If a question is not applicable to the project, please type "Nothing to report."

Grant Information	
Grant ID	549032
Award Program	Pollinator Health Fund
Project Title	LOCATION, LOCATION, LOCATION: DEVELOPING TOOLS FOR SELECTION AND MANAGEMENT OF LANDSCAPES TO PROMOTE HEALTHY BEE POPULATIONS
Grant Start Date	01/01/2018
Grant End Date	12/31/2021
Total Grant Budget	\$2,344,203
Total FFAR Budget	\$1,177,137.00
Matching Funder(s)	Matching funds: \$1,227,066.00
Final Report Date	04/25/22
Project Director/Principal Investigator Information	
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General Information

- 1.1. **Please list the geographic location(s) – city, state, congressional district - where the work was conducted. If the work was conducted outside of the US, please list the city and country.**

Penn State University: University Park, PA (12th district). Note that bees from feral honey bee colonies were collected from 27 sites in Pennsylvania, and bumble bees were collected from 22 sites across Pennsylvania.

Dickinson College: Carlisle, PA (11th district)

University of Minnesota: St. Paul, MN (4th district)

University of California Davis: Davis California (3rd congressional district).

- 1.2. **How many new jobs were created because of this grant funding?**

Total: 42

New: 26

Maintained = 16

Penn State University:

Eric Wafula (research assistant professor, 3 months), Paula Ralph (research tech, ~3 months) Natalie Boyle (research assistant professor, paid from PSU, match), Kate Anton (full time technician, paid from PSU, match), Heather Descorcie (educator, paid, FFAR), Holly Holt (part-time science communication, responsible for improving information on Beescape website and managing Beescape newsletter), Michele Mansfield (part time molecular technician supporting pollen DNA barcoding).

DJ McNeil (postdoc, paid 9 months from FFAR), Bogdan Caradima (postdoc, paid 5 months from FFAR), Alex Jordan (postdoc, paid 4 months from FFAR), Martina Calovi (postdoc, paid from other sources), Gabriela Quinlan (postdoc, paid from other sources), Doug Sponsler (postdoc, paid from other sources)

Timothy Prestby (PhD grad student, paid for two semesters), Chauncy Hinshaw (PhD grad student, paid for

summer), Elena Gratton (part time technician/grad student, paid from FFAR), Codey Mathis (part-time technician, graduate student, paid partly from FFAR, part from PSU; matching funds), Sarah Kern (undergraduate, paid from FFAR), Eric Homan (PhD student, 1 semester, Engineering), Tyler Jones, Briana Ezray (grad students, PSU funds, match), Makaylee Crone, Melanie Kammerer (contributed to research, not paid/match),

Undergraduate wages (Buck, Weber, Heimann, Kern)

Dickinson College: Undergraduate researchers Karan Shakya and Zoe Muller completed 8-week summer research experiences supported by this project. Undergraduate researchers Paige Baisley and Sara Soba also contributed to this project, but were paid from other sources.

University of Minnesota: These funds supported a postdoctoral research associate with matching funds and FFAR funds (Bruninga-Socular). A project manager (Michelle Vohs) was supported through this project. A PhD student (Brokaw) is working on this project but supported through other funds. Six undergraduate researchers were also supported from these funds. A research software engineer (Peter Hawthorne) spent one month of time with FFAR and matching funds to start development of a decision support tool with PI, Lonsdorf

University of California Davis: Uta Muller (Post doctoral Researcher), two undergraduate lab assistants (S. Perry, K Sobers-Harrington),

1.3. How many jobs were maintained because of this grant funding?

16

1.4. Have there been any changes to your organization's IRS 501(c)(3) non-profit status since you were awarded the grant? If yes, please explain.

NA

1.5. Has your organization undergone a recent name change? If so, please provide the new name of your organization.

NA

2. Scientific Report

2.1. Public Abstract (up to 500 words): The abstract should be 'stand-alone' and is intended for a general audience. It should be a concise overview/summary of the importance of the project, the issues that the project addressed, and the key findings of the project. The abstract

should also clearly state how the results of the project help to address important needs in U.S. food and agriculture systems.

Pollinators are in decline due to multiple, interacting stressors. Landscape factors can greatly impact the health of both wild and managed pollinators. In particular, diverse and abundant flowering plant species are both critical for providing high quality nutrition to pollinators, and can mitigate the negative impacts of other factors, such as pesticide or pathogen exposure. We will **(1)** determine what makes a landscape healthy for bees **(2)** develop tools to allow stakeholders to generate regionally-appropriate, cost-effective plant species mixes to meet different stakeholders' needs. **(3)** use information from Objectives 1 and 2 to develop an online decision support tool that will allow different stakeholders to a) assess the quality of their landscapes for supporting managed honey bee and wild bee populations and b) generate high-quality, cost-effective pollinator seed mixes to improve the forage quality of their landscapes. These studies are in collaboration with a diverse array of stakeholders, including beekeepers, growers, commercial plant producers, and conservation groups. This project bring together and leverages the expertise of outstanding teams of researchers, educators, and extension specialists from the northeastern, midwestern, and western United States.

The research, decision support tools, and stakeholder networks generated by this project will dramatically improve our understanding of the interacting factors driving losses of both managed and wild bees, and create the resources and partnerships needed to effectively mitigate these declines in a variety of landscapes across the United States.

- 2.2. Provide a description or interpretation of how the key findings of the project can be used by the agriculture community. It is important the PI identifies how these findings have been, or may be, adopted or adapted in U.S. agriculture and food systems. If the findings cannot yet be applied, this section should address how they can be used to guide future planning or decision-making (up to 1,000-word limit).**

These studies have improved our understanding of the landscape and climate factors impacting managed and wild bee health, and identify plant species and planting designs that best support pollinator habitat restoration and maintenance. This information has been incorporated into online decision support tools that will allow different stakeholders to a) assess the quality of their landscapes for supporting managed honey bee and wild bee populations and b) generate high-quality, cost-effective pollinator seed mixes to improve the forage quality of their landscapes.

- 2.3. Research Outcomes/Impact (up to 1,500-word limit). The primary goal here is to answer questions such as "How did this project lead to improvements in U.S. agriculture and food systems?" or "How can the findings of this study guide or make improvement in future investigations and research?" Outcomes should be explained and**

classified in one of the following ways:

- a) **potential outcomes, i.e., findings, results, or recommendations that could impact U.S. agriculture and food systems, if used;**
- b) **b. intermediate outcomes, i.e., how the findings, results, or recommendations have been used by others to influence practices, legislation, research/product design, training and so forth; and**
- c) **c. end outcomes, i.e., how findings, results, or recommendations have contributed to documented reductions in work-related morbidity, mortality, and/or exposure.**

These studies have improved our understanding of the landscape and climate factors impacting managed and wild bee health, and identified plant species and planting designs that best support pollinator habitat restoration and maintenance. This information has been incorporate into online decision support tools that will allow different stakeholders to a) assess the quality of their landscapes for supporting managed honey bee and wild bee populations and b) generate high-quality, cost-effective pollinator seed mixes to improve the forage quality of their landscapes.

2.4. What were the goals/specific aims of the project? If the approved application lists milestones/target dates for important activities or phases for this reporting period, identify these milestones and dates, as well as show actual completion dates or the percentage of completion of milestone targets. (up to 1,500-word limit)

OBJECTIVE 1. DETERMINE WHAT MAKES A LANDSCAPE ‘HEALTHY’ FOR BEES.

Objective 1.0. Continue to develop and refine indices of landscape quality (forage quality, pesticide use, nesting quality, climate severity).

Objective 1.1. Develop approaches for analyzing temporal data from automated hives scales.

Objective 1.2. Develop approaches for using existing data sets from USGS Bee Inventory and Monitoring Laboratory to evaluate how landscape impacts wild bee abundance and diversity.

Objective 1.3. Collect managed honey bees, feral honey bees, and bumble bees from across ~25 sites in PA that vary in their landscape indices and measure the extent of pathogen infections and immune challenge across each of these communities. Initiate analysis of viral loads in these samples using qRT-PCR.

OBJECTIVE 2. DEVELOP TOOLS ALLOWING STAKEHOLDERS TO GENERATE REGIONALLY-APPROPRIATE, COST-EFFECTIVE PLANT SPECIES MIXES THAT MEET DIFFERENT NEEDS.

Objective 2.1. Compile data on flowering phenology, pollinator phenology and pollinator preference and integrate these data into an existing seed mix analysis tool developed by PIs Lonsdorf and Williams to evaluate the cost effectiveness of existing mixes for supporting wild bees.

2.2. Quantify flowering phenology, pollinator phenology and pollinator preference among diverse climate-appropriate native and horticultural species.

Objective 2.3. Create a plant-pollinator interaction data clearinghouse that provides essential input for generating high-performing seed mixes across the US

Objective 2.4 Determine how seeding density and diversity influence plant establishment, floral area, and weed invasion.

Objective 2.5 Assess pesticide risk potential of various plant species based on likelihood of translocation of pesticides to floral resources.

OBJECTIVE 3. DEVELOP ONLINE DECISION SUPPORT TOOLS (DSTS) TO ALLOW STAKEHOLDERS TO EVALUATE THE QUALITY OF THEIR LANDSCAPES, BEEKEEPING PRACTICES, AND PLANTING REGIMES TO SUPPORT BEE HEALTH AND OBTAIN RECOMMENDATIONS FOR IMPROVEMENT.

Objective 3.1. Modify the Pollination Mapper DST for use by beekeepers, land managers and conservationists. Develop system to allow users to obtain information on their landscape quality, create individual accounts, and enter data on bee health metrics.

Objective 3.2. Create a simple interactive, online tool for land managers interested in developing restorations for bees.

- 2.5. **Have any of the major goals/specific aims or milestones for the project changed since the award? If so, please list the goal(s) that have changed and provide justification for the change from the approved goals. (up to 500-word limit)**

NA

- 2.6. **What was accomplished under the goals/specific aims or milestones for the project? Please describe in detail, 1) major activities; 2) specific objectives; 3) significant results, including major findings, developments, or conclusions (both positive and negative); and 4) key outcomes or other achievements. Include discussion of stated goals not met. The emphasis in reporting in this section should focus on accomplishments. In the response, emphasize the significance of the findings to the scientific field. This section can be as technical as the**

PI would like. (up to 5,000-word limit)

Objective 1.0.

Results of the insecticide toxic load index studies are published (Douglas et al, 2020), with a second manuscript describing methodology for finer-scale estimates in revision.

We collected data from the PA Bureau of Forestry on pesticide use patterns and tree species distributions (including flowering trees), to develop a more informative insecticide and forage map for wild bees in PA forests. Unfortunately the data was not comprehensive enough in time or space to allow for a detailed index.

We evaluated the weather and climate variables associated with managed and wild bee health (Calovi et al 2021, Kammerer et al 2021). The responses to these variables were quite distinct across bee species and thus we have not yet developed a generic “climate index” for bee health.

Objective 1.1.

We developed a workflow for analyzing honey bee colony weight data from Broodminder scales (Sponsler et al 2020). We analyzed data sets from studies conducted at Penn State, University of Wisconsin, Ohio State, Michigan State, USGS and University of Nebraska (Quinlan, McNeil et al in revision).

Objective 1.2.

Graduate student Kammerer organized, validated, and shared an analysis-ready version of the USGS Bee Inventory and Monitoring Laboratory data set (Kammerer et al 2021). She demonstrated that weather factors are the major drivers of wild bee species diversity and abundance in this data set (Kammerer et al 2021). Graduate student Eric Homan led an effort to develop a computational pipeline to automatically quantify insect abundance according to time and weather patterns using a camera tracking system, to improve our ability to monitor insect populations across diverse landscapes.

Objective 1.3.

We surveyed 25 pairs of feral and managed colonies over a 2-year period (Hinshaw et al 2021). Feral colonies had higher levels of DWV, and higher pathogen levels were associated with increased immune gene expression. Differential expression of key immune genes increased the odds of overwintering survival.

Pathogen loads in wild bumble bees in Pennsylvania were negatively associated with forage resource quality, positively associated with honey bee colony density, and influenced by latitude and longitude, suggesting potential impacts of weather (McNeil et al 2020). Our data suggest bumble bee species distributions vary with local habitat conditions. We collected samples to explore this in more detail in summer 2020. We sampled *Bombus impatiens* at 36 sites (12 forested, 12 edge, 12 valley) across central PA from June 22nd to July 8th 2020. There were no significant patterns associated with *Nosema bombi*; however we found that BQCV and DWV loads were highest in the valleys and lowest in the forests, while *Crithidia* loads were highest in the forests and lowest in the valleys. These data are being further analyzed and incorporated into a publication (Gratton et al in prep).

Objective 2.1.

The UMN team has nearly finished a manuscript, led by post-doc Bruninga-Socular, to evaluate the cost effectiveness of effort needed to develop plant-pollinator information necessary to run the seed mix model. They simulated seed mix designs resulting from reductions in effort (and cost) with four different approaches: reduced numbers of specimens sampled, reduced number of sites sampled, reduced number of sampling days per site and using representative taxon (bumblebees only). The team found that prioritizing broad spatial and taxonomic coverage of plant-bee interaction networks over intense local sampling in order to guide cost-effective seed mix design and bee monitoring.

Objective 2.2.

At PSU, we developed an optimized methodology and workflow for DNA metabarcoding of bee collected pollen, to identify the preferred floral forage plants (Sponsler et al 2020a,b). We collaborated with Professor Claude dePamphilis (PSU Biology) and Dr. Wafula to create a semi-automated bioinformatics workflow. We completed the collection and sequencing of wild bee collected pollen from spring and summer collections, but the data are still being processed. Preliminary results indicate that co-foraging spring-emerging bees are collecting from distinct flowering plants with different nutritional content, and forage on similar plants (primarily flowering trees) in both urban and natural areas in our study sites.

Objective 2.2.

In CA, the bee visitation data on 42 wildflower species collected over multiple years was analyzed in order to determine visitation profiles of individual plant species. Abundance of different pollinator groups as well as species richness of recorded native bees were summarized per plant. Additionally, plant-pollinator-interactions were investigated applying network analysis for early, mid and late seasonal networks. Species roles in the network based on centrality and specialization were analyzed in order to identify a species core that could provide stability to the whole network as well as individual species that could support specialized interactions. With data from the seed mix experiment, plant species were also evaluated for their performance in seed mix applications based on maximum floral area and phenology. The affiliation of those species with different network modules was characterized.

Differences of functional complementarity of native bee communities of the modules the selected species were associated with was tested. A set of plant species performing above average in seed mixes was found to be cores of different network modules. The network modules supported a species- and functionally diverse native bee community in the early but not in later seasons. The results provide insights for selecting “key plant species” as targets for pollinator restoration efforts.

In Minnesota, we planted 39 forb species in three monoplots each. We are combine visits from the monoplots with those of our plant by bee interaction network to determine which plants are the most visited by native bees across different phenological periods. We have also completed a list of the bees of Minnesota and have information on which of the 489 bee species are specialists and which plants these bee species rely upon. Specialists are often rare and of conservation concern. The information from the seed mix experiment enables us to create a seed mix that flowers across the entire season, will easily establish and will contain core flowering plants as well as plants that support bee species of conservation concern.

Objective 2.3.

In California we completed work on an additional 6 data sets on bee x plant interaction data sets and

phenological records from our studies with standardized pollinator habitat plantings. The UMN team compiled an interaction database with over 86,000 bee by plant interactions; the dataset is comprised of 531 plant species and over 450 bee species. These are ready for incorporation into the data clearinghouse. Cariveau and Williams are PIs on the NIFA funded U.S. National Native Bee Monitoring Research Coordination Network. This group continued to work toward strategies for national bee monitoring including prioritization of approaches and locations for efforts.

Objective 2.4.

We analyzed the data from the seed mix experiment for the 3 years. For CA, the data analysis revealed negative effects of grass inclusion over the 3 years on floral area and phenological coverage. A higher sown species richness resulted in a positive effect on phenology of seed mixes, particularly in the first years. Seed density did not impact either of the two tested variables and plant species richness was not affecting floral area. In total, the floral area of all plots was found to decline over the 3 years. In the first year there was distinct peak spring bloom which declined over the following years. Furthermore, there were less but still some floral resource provisions in the mid and late season in the first year which almost completely disappeared after the second year. The results of this study can help guiding land managers in design and applying pollinator seed mixes in different landscape contexts and conservation goals.

For MN, we finalized three years of data collection from a seed mix experiment that varied the number of seeded forb species, seeding density and grass to forb ratio. As seeded forb diversity increased, the total floral area and phenological coverage increased. In addition, plots with lower the forb to grass ratio experienced greater total floral area as well as phenological coverage. Interestingly, seed density did not affect floral area or phenological coverage. These effects were not found in the first year of the study but increased in subsequent years. This is consistent with anecdotal evidence that midwest tallgrass prairie restorations do not reach maturity for a 2-4 years past planting. In years 2 and 3, the late summer season had the greatest bloom due to goldenrod boom (*Solidago sp.*) Our results suggest that reducing the density of seeds planted will still provide abundant floral resources and using fewer seeds will reduce costs of seed mixes.

Objective 2.5.

To contribute to our understanding of potential pesticide risk through pollen and nectar routes of exposure and how this may vary among plant types, we ran a large field experiment at UC Davis. We selected replicate annual forb, and perennial forb/ shrub species that are commonly used in regional pollinator plantings (UC Arboretum Allstars). All plant species plants were grown in both container and inground field contexts that mimic the contexts in which plants in the garden and restoration trades will potentially be exposed to pesticides. Separate replicates of species in each growing context were treated with one of 4 different neonicotinoids (Imidacloprid, Thiamethoxam, Clothianidin, Dinotefuran). This created a fully replicated 3-way design with plant growth type, growing context, pesticide type. For each we exposed plants to the neonic according to label rates and timing (not during bloom) and then sampled pollen and nectar at subsequent time points after spray. Annuals were sampled at 15 and 30 day intervals, the longer-blooming perennials were sampled at 30 and 60 intervals. Samples have been sent to a certified California Department Food and Agriculture testing lab for chemical analysis. We are currently waiting for results to be released by CDFA testing lab. Analysis has been completed but remains sequestered by CDPR.

Objective 3.1.

For our Beescape interactive tool, from March 2021 to March 2022, over 11,000 individuals viewed the Beescape site, and the site has received over 16,000 views. We have 657 individuals subscribed to receive the Beescape newsletter, and the newsletter is also incorporated into the PA State Beekeepers monthly newsletter.

Graduate student Prestby worked with collaborator Professor Anthony Robinson (Penn State GeoVISTA Center) to evaluate usability and utility of the Beescape website. 30 beekeepers who had never used Beescape before were recruited for the user study, after IRB approval. Beekeepers found Beescape easy to use and informative but felt it could be more interactive and its predictions were not well-aligned with their expectation of local landscape quality. The manuscript describing this study is published (Robinson et al 2021).

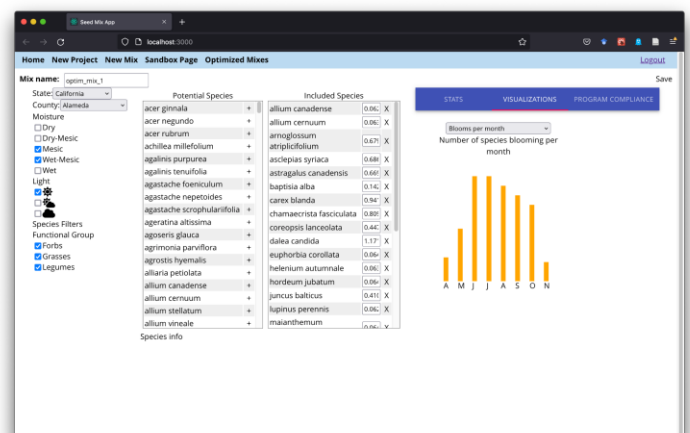
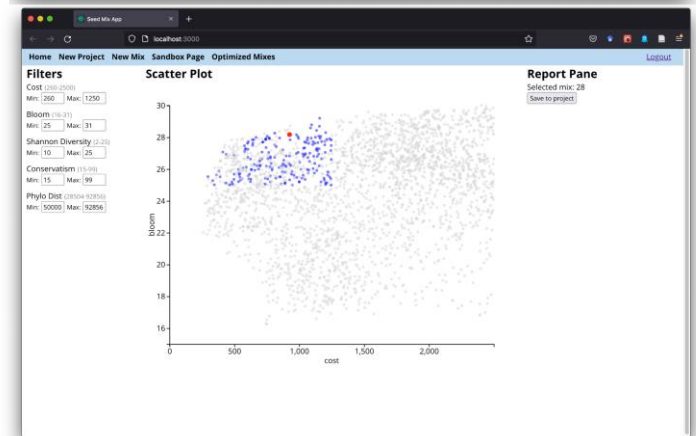
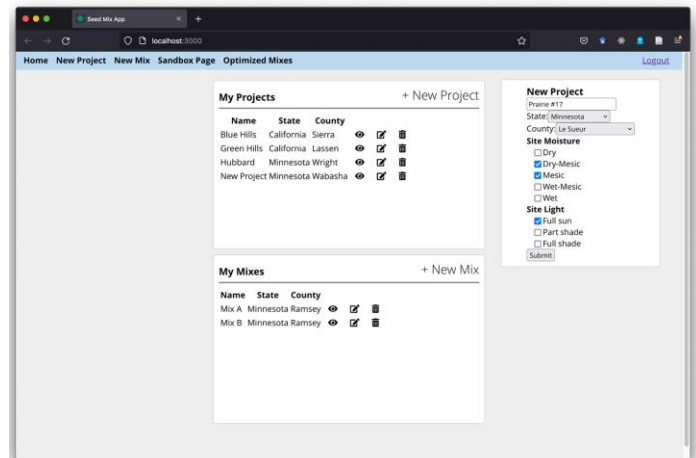
Objective 3. 2.

The MN team led development of a seed mix tool (web app, see figures) that integrates seed mix catalog information into the seed mix algorithm. The goal of the tool is to allow a user to save multiple mixes and reveal to the user ecological and evolutionary measures of diversity and support for pollinators. The user would first go to their homepage where their projects and mixes are organized. We imagine that a user might have several different restoration projects and consider multiple mixes at each project.

To initiate a new project, the user would set the context for the mix, describing the state, county and habitat for the restoration, e.g. mesic, dry, etc.

This step will filter out species that would not be appropriate for that location and habitat type.

Next, the user has two complementary options. They can simply start building their own mix from scratch or use the optimizer and let the multi-objective (pollinator coverage, cost, phylogenetic diversity, species diversity) optimization process create mixes. If the user chooses the latter, the tool provides a



viewer that allows the user to explore the results of the many mixes that result - those that are along a multi-objective pareto surface. The user can identify the set of mixes that meet their needs (blue highlighted) and then can select one of those mixes to export to their project (red). Once exported, the user can then view the mix and modify it as needed while seeing how the modified mix changes the metrics.

We will be continuing this work after the project funding ends to connect the web application with natural resource managers.

2.7. Discuss efforts taken to ensure the approach is scientifically rigorous. Include approaches taken to ensure robust and unbiased results.

All of our studies are designed to be published in the peer-reviewed scientific literature and thus are designed to be scientifically rigorous, robust and unbiased.

2.8. List key stakeholders that could be served by results of this project.

Beekeepers, growers, land manager and urban planners, conservation groups, homeowners

2.9. How have the results of this reporting period been disseminated to communities of interest? Describe how the results have been disseminated to communities of interest. Include any outreach activities that have been undertaken to reach members of communities who are not usually aware of such activities, to enhance public understanding and increase interest in learning and careers in science, technology, and the humanities. Reporting the routine dissemination of information (e.g., websites, press releases) is not required. For awards not designed to disseminate information to the public or conduct similar outreach activities, a response is not required; the grantee should write "nothing to report." A detailed response is only required for awards or award components that are designed to disseminate information to the public or conduct similar outreach activities. Note that scientific publications and sharing research sources will be reported under Information Products. (up to 1,000-word limit)

Below, we only highlight presentations from 2021. Please see previous reports for information from previous years.

We have discussed this project at multiple stakeholder conferences and K-12 classes/groups. These include:

- The Foundation for Food and Agricultural Sciences Annual Board Meeting (October 2021, Grozinger)
- American Honey Producers Association Annual Meeting (Boyle, November 2021)
- Pennsylvania State Beekeepers Association (November 2021, Lopez Uribe)
- Penn State Ag Council Annual Meeting (October 2021, Grozinger)
- The North American Pollinator Protection Campaign (October 2021, Grozinger)
- Girls Advancing in STEM (August, 2021, Grozinger and López-Uribe)
- Rotary Club of Langhorne, PA (May 2021, Grozinger)
- Beekeeping Today Podcast (April 2021, Grozinger, <https://www.beekeepingtodaypodcast.com/the-climate-change-impact-on-bees-floral-sources-with-dr-christina-grozinger-s3-e46/>)
- Radio Park Elementary School (April 2021, Grozinger, Boyle)
- Metro Atlanta Beekeepers Association (March 2021, Grozinger)
- Engineering Edge Podcast (April 2021, Grozinger; <https://theengineeringedge.podbean.com/e/the-everyday-tech-that-could-take-us-to-the-moon/>)
- STEM Professionals Panel for Chief Science Officers Program (January 2021, Grozinger).
- Beekeepers of ABCI Mid-Winter Seminar. Kittanning, PA (August 2021, Quinlan)
- PA State Beekeepers Association Annual Meeting and Conference. State College, PA (November 2021, Quinlan)
- Almond Conference, CA (December 2021, Niño)
- California State Beekeepers Association (November 2021, Niño)
- Finding genius Podcast (October 2021, Niño)
<https://www.findinggeniuspodcast.com/podcasts/apiculture-and-working-with-honeybees-to-improve-queen-bee-and-colony-reproductive-health-with-elina-l-nino/>
- Oxford Climate Review Interview (October 2021, Niño)
- San Diego Beekeeping Society, CA (October 2021, Niño)
- New Roads School (4th grade interview), CA (September 2021, Niño)
- CNN Interview – Impact of Drought on Bee Health (June 2021, Niño)
- Santa Clara Bee Guild, CA (June 2021, Niño)
- Humboldt County Beekeepers Association, CA (May 2021, Niño)
- El Dorado County Master Gardeners, CA (May 2021, Niño)
- Seed Production Course, UC Davis, CA (February 2021, Niño)
- Environmental Defence, Non-cropped Land use options for pollinators and beneficials (UC Davis, virtual (Williams July 28 2021
- Silicon Valley AgTech Forum,, Seeds for our future, Design and placement of habitat for pollinators (Williams, August 19, 2021)
- KIND Foundation, Designing habitat for multiple benefits to bees and pollination (Williams, October 19, 2021)
- California Pollinator Coalition Meeting Challenges and solutions for bees in CA agriculture (Williams, February 24, 2021)

We discussed this project at multiple scientific conferences and institutions. These include:

- “The Larry Larson Symposium: Scientific Advances on Insect Species Adaptation to the Impact of Climate Change and Habitat Transformation” Entomological Society of America Annual Meeting. Denver, CO (October 2021, Grozinger).
- “From flowers to landscapes: the natural processes influencing pollinator health” Kew Royal Botanical Gardens symposium, (October 2021, Grozinger)
- Pollination Conservation and Ecology Webinar Series. US FWS National Conservation Training Center. Online (June 2021, Grozinger)
- Mainstreaming Biodiversity in a Decade of Action. Penn State Sustainability Institute virtual webinar series. (February 2021, Grozinger)
- USDA-ARS Big Data in Agricultural and Pollinator Research Workshop. Virtual (August 2021, Quinlan)
- Tradeoffs of adapting to a changing world. Entomology Society of America Annual Meeting. Denver, CO (October 2021, Quinlan)
- Center for Pollinator Research Symposium. State College, PA (November 2021, Quinlan)
- “Pollinators, pesticides, and people: The socioecology of insect declines.” University of California Riverside, California Agriculture and Food Enterprise seminar series. Virtual. (May 2021, Douglas)
- “Supporting pollinators in California agricultural and urban environments”, Entomological Society of America Pacific Branch Meeting (April 2021, Niño)
- Wild flower seed mixes for native bee habitat enhancements in agricultural landscapes”, Entomological Society Denver (October 2021, Williams, Uta Müller)
- Pollinator habitat enhances nesting opportunities for wild bees, Entomological Society Denver (October 2021, Williams)
- Orchard Bee Association Meeting Effects of resource and pesticide stressors on bees (Williams December 11, 2021)

Additional outreach programming for the public developed at Penn State by education specialist Heather Desorcie:

- Developed lists and pamphlets of high attraction pollinator plants and life histories of the five main pollinator taxa (focused on the mid-Atlantic and Northeastern US) for a wide audience, including gardeners, classroom teachers, and the general public.
- In-person educational programs have been developed, facilitated and offered, including using the Pollinator and Bird Garden at The Arboretum at Penn State as a living classroom. Program topics include solitary cavity-nesting bee life history, bee hotel management, honey bee behavior in springtime, importance of honey bees in agriculture, and choosing plants for specific pollinator taxa.
- Created a robust social media plan for Facebook, Instagram, and Twitter. In addition to disseminating research results, our social media presence encourages viewers to explore, understand, and promote insect conservation effort,

Additional outreach programming for the public developed at UC Davis by extension specialist Elina L. Niño and California Master Beekeeper Program staff:

- Virtual and in-person programming has been developed and conducted, including basic beekeeping courses and workshops for using in-hive technologies such as Broodminder scales to track changes in honey bee colony parameters in response to environmental inputs (UC South Coast Research and Extension Center, CA Master Beekeeper Apiary).

2.10. Describe challenges or delays encountered during project and actions that were taken to resolve them. Only describe significant challenges that may have impeded the research and emphasize their resolution. (up to 500-word limit)

As a result of COVID-related restrictions to field and lab research, we applied for a no-cost extension to extend the original grant from 12/31/2020 to 12/31/2021.

2.11. What opportunities for training and professional development has the project provided during this reporting period? If the research is not intended to provide training and professional development during this period, state "Nothing to Report." For all projects reporting graduate student and/or post-doctoral participants, grantees are encouraged to describe how Individual Development Plans (IDPs) are used to help manage the training for those individuals. (up to 500-word limit)

Penn State University: All individuals affiliated with this project receive integrative training in pollinator biology, health and ecology which spans genomics to geospatial analyses. Postdoctoral fellow DJ McNeil took a lead role in assessing landscape effects on bumble bee health in Objective 1.3, and managing the Beescape Platform (Objective 3.1). McNeil has accepted an assistant professor position at UNC Wilmington, to start in July 2021. Postdoctoral fellows Bogdan Caradima (paid from this project), Gabriela Quinlan and Martina Calovi (affiliated with the project but paid from other sources) developed new skills in multivariate statistical analyses. Calovi accepted an associate professor position in Norway and Quinlan received a NSF postdoctoral fellowship, while Caradima accepted a position at an environmental monitoring company. Dr. Eric Wafula led the development of a semi-automated bioinformatics pipeline for DNA barcoding analysis of pollen, and accepted a bioinformatic position in the health care industry in April 2021. Graduate student Chauncy Hinshaw (who was paid by FFAR) was awarded a USDA NIFA Predoctoral Fellowship. Graduate student Melanie Kammerer, who was affiliated with this project but supported by a USDA predoctoral fellowship, graduated in 2020 and is currently a USDA SCINET fellow. Graduate student Timonhy Prestby is receiving training in geospatial analyses, design and data visualization, and secured an internship with Apple Maps for the summer 2021, and has entered the PhD program at Penn State, and just received an NSF GRFP. Graduate student Eric Homan is defending his thesis in Summer 2022 and will be seeking a position in computer science industry. Former graduate students (paid by matching funds) Tyler Jones graduated with a MSc and works in science communication at NPR, while Briana Wham is a tenure track STEM library faculty at Penn State. Graduate student Makaylee Crone (affiliated with the project but is supported by an NSF GRFP), has the opportunity to learn molecular biology, genomics and bioinformatics to study plant-pollinator interactions. Technician Elena Gratton conducted wild bumble bee sampling with McNeil in 2020, and joined the Entomology MS graduate program (co-advised by PIs Hines and Grozinger) in fall 2021. Codey Mathis joined the project in

Spring 2020 as a part-time technician and joined the Entomology PhD graduate program in fall 2020 (coadvised by PIs Patch and Grozinger), and together with undergraduate Sarah Kern developed skills in monitoring and evaluating plant-pollinator interactions.

UC Davis: Two field assistants, Serra Perry and Kaleb Sobers-Harrington were involved in data collection, experimental set-up maintenance and data entry and worked with and were mentored by postdoc Uta Muller. An additional undergraduate student Li Wang, completed senior research project analyzing network data from our pollinator plots. He gained skills in network analysis using R and in scientific writing. Postdoc Uta Muller got trained in project management, data analysis with a focus on plant-pollinator networks as well as supervision of field assistants. Furthermore she visited the partner project in Minnesota for a common data analysis work project.

At UMN, [Bethanne Bruninga-Socular](#) was a postdoctoral research associate and managed much of this research. She increased her skills in experimental field studies, data management, and data analysis. A key part of her role was to mentor undergraduates and this led to presentations at a national meeting (see below). Julia Brokaw, a PhD student, was involved with the design, data collection and analysis of this project. She presented this research at the Society for Ecological Restoration Midwest Great Lakes Chapter Meeting. She has participated in the writing of manuscripts and will be a co-author on at least two manuscripts currently near completion. She was also acted as a mentor to undergraduate students on the project. Brokaw will be defending her dissertation in fall 2022. Six undergraduate technicians have been integral to completing this research. In 2020, three of these technicians presented the results from independent projects from this project at the Entomological Society of America national conference.

- 2.12. Please indicate the number of undergraduate and graduate students, post-doctoral scholars, or other educational components involved during this reporting period. If other education components are involved, please describe them in detail. (up to 500-word limit)

Undergraduates: 9
Graduate Students: 10
Postdoctoral Fellows: 7

3. Information Products

- 3.1. **Please list the type(s) of information products (e.g., scholarly publications, reports or monographs, workshop summaries or conference proceedings, video, audio, images, models, software, curricula, instruments or equipment, intervention, etc.) produced**

during the project resulting directly from the FFAR award.

Scholarly publications
Website

3.2. Please provide a list of citations for the information products produced during the project.

Robinson, A.C., Peeler, J.L., Prestby, T., Goslee, S.C., Anton, K., and C. M. Grozinger. “Beescape: Characterizing User Needs for Environmental Decision Support in Beekeeping” *Ecological Informatics* 64: 101366 <https://doi.org/10.1016/j.ecoinf.2021.101366> (2021).

Calovi, M., Grozinger, C., Miller, D., Goslee, S. “Summer weather conditions influence winter survival of honey bees (*Apis mellifera*) in the northeastern United States” *Scientific Reports* 11: 1553 (2021) <https://doi.org/10.1038/s41598-021-81051-8>

Kammerer, M., Goslee, S., Douglas, M.R., Tooker, J.F., Grozinger, C.M. “Wild bees as winners and losers: relative impacts of landscape composition, quality, and climate.” *Global Change Biology* January 12 <https://doi.org/10.1111/gcb.15485> (2021)

McNeil, D.J., McCormick, E., Heimann, A., Kammerer, M., Douglas, M., Goslee, S.C., Grozinger, C.M., and H. M. Hines. “Bumble Bees in Landscapes with Abundant Floral Resources Have Lower Pathogen Loads”. *Scientific Reports* 10(1), 1-12 (2020)

Kammerer, M., Tooker, J.F. and C.M. Grozinger. “A long-term dataset on wild bee abundance in Mid-Atlantic United States” *Scientific Data* 7, 240. <https://doi.org/10.1038/s41597-020-00577-0> (2020)

Douglas, M.R., Sponsler, D.B., Lonsdorf, E.V. and **C.M. Grozinger**. “County-level analysis reveals a rapidly shifting landscape of insecticide hazard to honey bees (*Apis mellifera*) on US farmland” *Scientific Reports* 10(1), 1-11 <https://doi.org/10.1038/s41598-019-57225-w> (2020).

Hinshaw, C., Evans, K.C., Rosa, C. and López-Urbe, M.M., “The role of pathogen dynamics and immune gene expression in the survival of feral honey bees.” *Frontiers in Ecology and Evolution*, 8, p.505. (2021)

3.3. Are there publications or manuscripts accepted for publication in a journal or other publication (e.g., book, one-time publication and monograph) during the project resulting directly from the FFAR award? If yes, please provide citation.

Douglas, M.R., Baisley, P., Soba, S., Kammerer, M.A., Lonsdorf, E.V., and C.M. Grozinger. “Putting

pesticides on the map for pollinator research and conservation” *Scientific Data* (in revision; pre-print available on bioRxiv: <https://www.biorxiv.org/content/10.1101/2021.10.18.464808v1.full>)

Quinlan, G.M., Sponsler, D., McNeil, D.J., Gaines-Day, H.R., McMinn-Sauder, H.B.G., Otto, C.R.V., Smart, A.H., Colin, T., Gratton, C., Isaacs, R., Johnson, R., Milbrath, M.O., and C. M. Grozinger . “Cool, dry climates and low precipitation lead to higher rates of summer forage availability and intake by honey bees (*Apis mellifera*) in the Northeast and Midwest, USA” *Environmental Research letters* (in revision).

Niño, E. L. , Yokota, S. Stacy, W. O. H., and Arathi, H. S. “Dietary phytochemicals alter hypopharyngeal gland size in honey bee (*Apis mellifera* L.) workers” *Heliyon* (In review).

3.4. Website(s). List the URL for any internet site(s) that disseminates the research activities. A short description of each site should be provided. It is not necessary to include the publications already specified above.

The Beescape portal (beescape.org) serves as a decision support tool for beekeepers and others to evaluate the quality of their landscapes for supporting bee populations. It also allows beekeepers to provide data on the health of their bees.

Penn State’s Center for Pollinator Research (<https://ento.psu.edu/pollinators>) and Insect Biodiversity Center (<https://www.huck.psu.edu/institutes-and-centers/insect-biodiversity-center>) websites includes press releases and other information.

Blog: update about feral bee project: <https://lopezuribelab.com/2018/09/11/tracking-health-feral-bees-pa-2018/>

Blog post: How viruses, immunity, and overwintering survival of feral honey bees compare to managed colonies
<https://lopezuribelab.com/2021/01/13/tracking-feral-bee-health-how-viruses-immunity-and-overwintering-survival-of-feral-honey-bees-compare-to-their-managed-neighbors/>

Insecticide Explorer: <https://insecticideexplorer.shinyapps.io/insecticideexplorer/>
This website was developed to provide access to crop-specific trends in insecticide use and intensity, translated into units of bee toxicity, for researchers and stakeholders.

3.5. Have inventions, patent applications, and/or licenses resulted from the project (U.S. and international)? If yes, indicate the invention, patent application(s), and/or license(s).



NA

- 3.6. **Are any of the information products produced during this grant that are confidential, proprietary, or subject to special license agreements? If so, please list them below and describe why they must remain confidential. Also, note if (and when) you plan to make these data publicly available in the future or if they must remain confidential indefinitely. (up to 500-word limit)**

NA

- 3.7. **Beyond depositing information products in a repository, what other activities have you undertaken to ensure that others (e.g., researchers, decision makers, and the public) can easily discover and access the listed information products? What other activities have you undertaken to ensure that other can access and re-use these data in the future?**

- Information related to this project is included in monthly newsletters distributed to Beescape participants and members of the Pennsylvania State Beekeepers Association.
- Information related to this project is included in weekly newsletter distributed to the Center for Pollinator Research and members of the Pennsylvania Pollinator Protection Campaign.
- The Beescape portal is now available for beekeepers and others to use. Information about this portal has been included in numerous press releases, including from the USDA.

4. Data Management

- 4.1. **Did the project generate any data? Data generation includes transformation of existing data sets and data from existing resources (e.g., maps and imageries). Please list the data generated for the award.**

- R code for generating raster files containing landscape quality scores for forage, nesting, and pesticide is available at [land-4-bees/PollinatorLandscapesInMatlab](#).
- R code for geospatial analyses and pollen DNA metabarcoding are housed in a private group GitHub folder.

- Data from beekeepers is entered in Beescape has been downloaded and preserved on hard drives in the Grozinger Lab
- Matlab Code for seed mix analyses currently held in a Google Drive Folder.
- A complete data set on wild bee abundance and diversity was made publicly available here: Kammerer, M., Tooker, J.F. and C.M. Grozinger. “A long-term dataset on wild bee abundance in Mid-Atlantic United States” *Scientific Data* 7, 240. <https://doi.org/10.1038/s41597-020-00577-0> (2020)
- Data used to analyze responses of wild bees to climate and land use is publicly available with the publication: Kammerer, M., Goslee, S., Douglas, M.R., Tooker, J.F., Grozinger, C.M. “Wild bees as winners and losers: relative impacts of landscape composition, quality, and climate.” *Global Change Biology* January 12 <https://doi.org/10.1111/gcb.15485> (2021)
- Data and code associated with the insecticide index will be made available through here: Douglas, M.R., Baisley, P., Soba, S., Kammerer, M.A., Lonsdorf, E.V., and C.M. Grozinger. “Putting pesticides on the map for pollinator research and conservation” *Scientific Data* (in revision; pre-print available on bioRxiv: <https://www.biorxiv.org/content/10.1101/2021.10.18.464808v1.full>)
- Data and code associated with our county-level pesticide analysis is publicly available with the open-access publication: Douglas, M.R., Sponsler, D.B., Lonsdorf, E.V. and **C.M. Grozinger**. “County-level analysis reveals a rapidly shifting landscape of insecticide hazard to honey bees (*Apis mellifera*) on US farmland” *Scientific Reports* 10(1), 1-11 <https://doi.org/10.1038/s41598-019-57225-w> (2020).
- Data from the three years seed mix experiment was entered into the Williams Lab database
- Data on plant-pollinator visitation on monoplots of native wildflowers from a total of 5 years was cleaned and prepared for network analysis
- Data from the seed mix experiment is currently on the Cariveau Lab MySQL database.
- The bee by plant data are stored on the Cariveau Lab MySQL database.
- The seed mix plots in Minnesota will continue for at least two more years as Cariveau secured more funding from the state of Minnesota. The new research will be to study plots long term. We will be collecting and storing the new data on the Cariveau Lab MySQL database and will acknowledge FFAR on future publications as this funding was integral to getting the research started.

4.2. **If you list multiple data sets, are these data sets related? If so, please provide a short description of how they are related.**

NA

4.3. **Please provide copies of relevant metadata records to support FFAR’s mission of enhancing the discoverability of FFAR funded project data and information products. Include or attach copies of records and a**

simple file inventory, if necessary, in a compressed folder.

We can share these data if FFAR would like, but most of the relevant data sets are already publicly available, including standardized, machine-readable metadata.



Philip Frum

Certification

The undersigned hereby certifies that to the best of their knowledge and belief, the above report and all supporting documents are true, accurate and complete. I am aware there is a significant penalty for submitting false or misleading information.

If applicable, I agree that my electronic signature is legally binding, equivalent to, and has the same validity and meaning as my handwritten signature. I will not claim otherwise.

PI Full Name: Christina Grozinger

PI Signature: *Christina Grozinger* Date: April 16, 2022

Authorized Signing Official (ASO) Name: Philip Frum_____

ASO Signature: *Philip Frum* Date: 4/25/2022